Exp. No:09

Implementation Of Subnetting In Cisco Packet Tracer Simulator

Aim:

To design multiple Subnets, assign static IP address connects in cisco packet tracer.

Objective:

* To create multiple subnet with a suitable number of hosts.
* To assign static IP addresses to all devices.
* To connect different subnet using a router.
* To test communication between subnet with and without a router.

Requirements:

* Cisco Packet Tracer.
* 1 router.
* 2 switches.
* 4 pcs.

Procedure:

a) Design multiple Subnet with Suitable number of hosts:

* Create two subnet using cisco packet tracer with 1 router, 2 switches and 4 pcs.
* Connect two pcs to each switch and connect both switches to the router.

⇒ Subnet 1 : 192.168.1.0/26

⇒ Subnet 2 : 192.168.1.64/26

b) Assign statics IP addresses across all subnets and connect the subnets via router.

Device	IP Address	Subnet mask
PC_0	192.168.1.2	255.255.255.192
PC_1	192.168.1.3	255.255.255.192
PC_2	192.168.1.66	255.255.255.192
PC_3	192.168.1.67	255.255.255.192
Router Gig0	192.168.1.1	255.255.255.192
Router Gig1	192.168.1.65	255.255.255.192

c) Simulate packet transmission across the subnet and observe the result.

* Use the ping command to test communication between PCs in different subnets.

Ping 192.168.1.66

OBSERVATION:

a) When subnets are connected via a router:

* Packets are successfully transmitted between the two subnets. The router forwards packets to the destination network, and the ping command receives replies.

b) When subnets are not connected via a router:

* PCs in the same subnet can communicate through the switch. PCs in different subnets cannot communicate because there is no router to forward packets between them.

RESULT:

Multiple subnets were designed in Cisco Packet Tracer, and static IP addresses were assigned to all devices. Communication between subnets was tested with and without a router.